

Elephant Herd: Partitioning a Cached Corpus Across Listening Agents

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September 30, 2026

Abstract

Keeping an entire corpus resident in a language model’s context, rather than retrieving fragments of it per query, was until recently priced out of existence. Two independent cost reductions changed that: capable small models, roughly $67\times$ cheaper per token than frontier models, and prompt caching, a further $5\times$ on repeated reads. Together a $334\times$ reduction, which takes a turn over a 1.19M-token corpus from \$5.95 to \$0.018.

We describe **Elephant Herd**, which partitions a corpus across N persistent listeners, each holding its partition as a cached prefix, and broadcasts every conversational turn to all of them. Each listener contributes a cited fact or stays silent, and a leader merges what returns into a briefing.

Measured on RULER with seven listener models and on a test built from GitLab’s public handbook, four results. **Partitioning does not improve retrieval**: a 7-way herd and a single model reading everything both score 100% at 131K, 256K and 512K tokens, at the same cost; the case for partitioning is capacity and specialisation, not recall. **Partitioning does not break chained facts either**, provided each listener is asked for relevant material rather than for the answer: split seven ways, one round scores 9 or 10 out of 10 for five of seven models at 128K, and DeepSeek-V4.1-Flash stays at 10 of 10 up to 512K. **Cheap listeners match a frontier one**: four models costing \$0.004 to \$0.086 per query score the same as Claude Sonnet 5 at \$0.357. **On realistic questions the herd answers 111 of 116**: 96 questions asked in other words than the text, plus 20 overheard conversations, over eight sections of the GitLab handbook (806K tokens), for \$0.59 per full run, against 95 for a standard retrieval pipeline given the same questions, model and grader. We also report how an earlier version of our own harness produced 0% on the chain task for all seven models, Sonnet included, and why a uniform zero across vendors should be read as a fault in the harness before a property of the method.

1 Motivation

A project folder grows for months. The assistant working in it sees whatever a retriever fetched for the current question, and nothing else. A decision written down in March is never read again.

This looks like a retrieval problem and is usually treated as one. We suggest it is a different problem. Retrieval runs when somebody asks a question, and most of the time nobody asks. A note about pricing does not surface during a conversation about pricing unless someone stops and goes looking for it.

Elephant Herd inverts this. Agents hold the notes in context and observe the conversation. When one recognises something relevant, it says so. Nobody has to ask.

What makes this possible now is not a new algorithm. Context stopped being expensive.

The cost argument only holds if a cheap listener is adequate for the task. We measure that directly in Section 3.3, rather than assuming it from price or from published long-context scores.

Configuration	Cost per turn, 1.19M-token corpus
Frontier model, uncached	\$5.95
Frontier model, cached	\$0.60
Small model, uncached	\$0.089
Small model, cached	\$0.018

Table 1: Two effects multiply: model choice ($\sim 67\times$) and caching ($\sim 5\times$). The first made a resident corpus viable. The second made it cheap to read repeatedly, which is what a listener hearing every turn needs.

2 System

A *herd* is N *elephants* plus a *leader*.

Each elephant holds one partition as a cached prefix. Partitions follow domain boundaries wherever possible, splitting only when a domain exceeds the context window; assembly is byte-deterministic so the cached prefix stays valid across turns.

Every turn is broadcast to all elephants. A local turn gate drops conversational filler before it reaches them, which costs nothing and removes a class of false contribution: an elephant cannot speak on a turn it never received.

Each elephant replies with a cited contribution or the literal token `<silent/>`. Silence is the expected case.

The leader merges contributions into a briefing for the calling agent. The audience is a model rather than a person, so the output is facts with source paths and no prose. When nothing was contributed it emits nothing. When two elephants disagree it reports both rather than picking one.

An optional iterated mode (`--deep`) re-broadcasts the findings so far, so an elephant can extend what another found. None of the results below needed it.

2.1 Implementation notes that turned out to matter

Provider pinning is experimental control, not cost hygiene. The same model is served by roughly 30 hosts at cache-read prices spanning $10\times$ and quantizations from fp4 to bf16. Unpinned routing varies recall between calls.

Listed cache prices are unreliable. One provider billed $4.2\times$ its advertised cache-read rate, because a five-minute TTL means calls inside the window re-create the cache while still reporting a 100% cache hit. Another reported a 100% hit on an 806K-token prefix and billed \$0.037 for the turn, against \$0.040 uncached; a third host serving the same model billed \$0.0037.

Truncation is indistinguishable from silence. Reasoning tokens come out of the same budget as the reply. A listener that exhausts that budget while reasoning returns an empty string, which looks exactly like a deliberate `<silent/>`. Unless the finish reason is checked, a budget problem is scored as a recall failure. Section 3.4 shows this happening at a 2,000-token cap.

Ask listeners for material, not answers. A listener told to answer or stay silent will correctly stay silent when its partition holds only part of an answer, and the part never reaches the merge. Section 3.4 measures the effect.

Reasoning must be enabled. Disabling it saves around 200 output tokens per silent turn, worth \$0.0001 against a \$0.0106 cache read, and loses correct answers.

3 Experiments

RULER [1] (Apache 2.0). **single** is one call over the full context; **herd** is 7 calls over a seventh each, reading the same total tokens, with the replies concatenated and scored by RULER’s rule that every gold string must appear. Every listener is pinned to one host, so routing cannot vary quantization or recall between calls.

Sections 3.1 and 3.2 use GLM-5.3-Flash on haystacks generated from Paul Graham essays. Sections 3.3 to 3.5 use seven models on the public RULER set published on Hugging Face as **self-long/RULER-llama3-1M**, $n = 10$ per cell. Costs are OpenRouter prices in September 2026 for one uncached query of about 128K input tokens; token counts are given so they can be recomputed as prices change.

3.1 Retrieval is preserved

Task	Single	Herd	Mean speakers
Single needle	100%	100%	1.0 / 7
Needle among distractors	100%	100%	2.9 / 7
Four scattered values	100%	100%	3.8 / 7

Table 2: $n = 10$ per task at 131K tokens.

The multi-value case is the notable one: the four gold values are scattered, scoring requires all of them, and roughly 3.8 elephants spoke per sample, so the values genuinely landed in different partitions and the merge recovered them. Silence discriminated cleanly, with zero false contributions across 70 listener calls on the single-needle task.

3.2 Context length: no separation up to 512K

If partitioning helped recall, the advantage should appear as the single context grows while each partition stays small. It does not.

Context	Single	Herd	Cost, single	Cost, herd
131K	100%	100%	\$0.0072	\$0.0055
256K	100%	100%	\$0.0121	\$0.0126
512K	100%	100%	\$0.0219	\$0.0235

Table 3: $n = 8$ per length. No degradation, so the curves never diverge.

Partitioning does not improve retrieval at any length we can test. The reasons to partition are capacity and specialisation: a corpus larger than any context window has no single-context arm at all, and partitions mapping to real domains make “which specialist should have known this?” answerable, which is what silence and attribution depend on. Neither is expressible in RULER, whose haystacks are arbitrary essay chunks.

3.3 Chained facts survive partitioning

Variable tracking asks for every variable that ends up holding a value, where the value is assigned once and then passed along a chain: **VAR A = 46208**, **VAR B = VAR A**, and so on, five variables in

all. The links are scattered through 128K tokens of filler, so with seven partitions no listener sees the whole chain. This has the same shape as tracing a symbol through a codebase or following a decision through notes that refer to each other: the answer is written down nowhere, and has to be assembled.

Each listener received its partition, the question, and this instruction:

You hold part of a longer text. Report anything in your part that is relevant to the question, quoted exactly. Include things that only point at something else instead of answering it. Others hold the rest of the text, so do not try to work out the final answer. Nothing relevant? Reply: `<silent/>`

with a reply budget of 20,000 tokens, one round, no iteration.

Listener	One fact	Several facts	Chain	Cost per query
DeepSeek-V4.1-Flash	10/10	10/10	10/10	\$0.011
Gemini 3.7 Flash	10/10	10/10	10/10	\$0.086
Claude Sonnet 5	10/10	10/10	9/10	\$0.357
GLM-5.3-Flash	10/10	10/10	9/10	\$0.016
Qwen3.7-Flash	10/10	10/10	9/10	\$0.004
Xiaomi MiMo-v2.5	10/10	8/10	2/10	\$0.012
Gemini 2.5 Flash-Lite	10/10	9/10	0/10	\$0.016

Table 4: Herd of 7, 128K tokens, one round. “One fact” is RULER `niah_single_2`, “several facts” `niah_multivalue`, “chain” `vt`. Cost is for the chain task.

Five of seven listeners recover the chain in a single round, so partitioning does not break linked questions. A single call over the whole context scored 3 of 3 (GLM-5.3-Flash) and 2 of 2 (DeepSeek-V4.1-Flash) on the same data, so the herd matches the unpartitioned case; that comparison is small and was run under the old instruction below, which suits a single reader since it can see the whole chain.

The listeners that succeed include four whose cost per query is between a fourth and a ninetieth of Sonnet 5’s, with the same or one more correct answer. Price did not predict the two failures, and neither did the other two tasks: Gemini 2.5 Flash-Lite is near perfect on one fact and several facts and scores zero on the chain. Both failing models also fail under the old instruction below, so the change of instruction does not explain them; we have not established what does.

3.4 A harness fault that looked like a finding

An earlier version of this paper reported that partitioning destroys chained facts: 0% for the herd against 100% for a single model, recovered only by iterating the herd at about three times the cost. That result came from two choices in our harness, not from partitioning.

The first was the instruction. Listeners were told to answer the question from their part, or reply `<silent/>`. A listener holding `VAR LKDVP = VAR TWGEN` has no answer to “which variables equal 46208?”, since its line holds no value, so silence was the correct reply and every listener gave it. The second was a reply budget of 2,000 tokens, which reasoning could exhaust before any answer was written, returning an empty string scored as silence.

In a smaller run ($n = 3$), changing the instruction alone lifted Sonnet 5 and Gemini 3.7 Flash to 3 of 3 but the three cheapest models only to 1 of 3; adding the larger budget lifted the two cheap models we retested to 3 of 3. Both changes were needed.

Listener	Old: answer or silent, 2K	New: report relevant, 20K
DeepSeek-V4.1-Flash	0/10	10/10
Gemini 3.7 Flash	0/10	10/10
Claude Sonnet 5	0/10	9/10
GLM-5.3-Flash	1/10	9/10
Qwen3.7-Flash	0/10	9/10
Xiaomi MiMo-v2.5	0/9	2/10
Gemini 2.5 Flash-Lite	0/10	0/10

Table 5: Chain task, same data and settings apart from the instruction and the reply budget. One Xiaomi sample was lost to a provider error. Under the old settings the one fact and several facts tasks scored 9 or 10 out of 10 for every model, so the fault was specific to answers spread across partitions.

Two lessons generalise. A listener that follows a strict silence instruction faithfully hides partial information the system needs, while one that ignores it is rewarded; a benchmark built that way measures obedience. And seven models from six vendors scoring zero on the same task is more likely a fault in the harness than a property of the method. We read it as the latter for several weeks.

Results from the earlier version that depended on this harness are withdrawn: the 0% single-round chain result, the scaling of iteration rounds with chain length, and a finding about the rule for halting iteration. Iteration was compensating for the harness, and is not needed for any result reported here.

3.5 Chains at 256K and 512K

Splitting 512K tokens seven ways gives each listener about 73K. We repeated the chain task at 256K and 512K with one model reading everything and with the herd, for the two cheapest listeners, $n = 10$.

Listener, instruction	128K	256K	512K
DeepSeek-V4.1-Flash, single call	10/10	10/10	10/10
DeepSeek-V4.1-Flash, herd	10/10	10/10	10/10
GLM-5.3-Flash, single call	10/10	10/10	10/10
GLM-5.3-Flash, herd	9/10	7/10	1/6
GLM-5.3-Flash, herd, “report, not judge”	–	9/10	8/10

Table 6: Chain task. GLM at 512K lost four samples to host errors. DeepSeek rows use the second instruction and a 60K reply budget; GLM’s first herd row uses the instruction of Section 3.3 and a 20K budget.

GLM’s reasoning on a silent reply shows why it dropped links: it found its line (VAR AMBYO = VAR COWQQ), read the instruction to include pointers, and decided the pointer was irrelevant because the other end was not in its part. The same slice asked again sometimes reported the line and sometimes did not. Replacing the instruction with one that says the listener’s job is to report, not to judge, raised GLM from 1 of 6 to 8 of 10 at 512K. More reply budget did not raise it further, and a separate cap on reasoning at 8K left accuracy unchanged while cutting cost by a third and latency fivefold. DeepSeek needed neither change.

3.6 Realistic questions over a real handbook

RULER hides one fact in filler and asks with the text’s own words. Real notes differ: many documents on one subject, answers that change over time, more than one right answer, and questions in other words. We built a test from the public GitLab handbook, pinned to commit [7a2e264](#): eight sections (product, finance, total rewards, leadership, hiring, communication, values, CEO), 806K tokens, one listener per section, each holding its section as a cached prefix through the same code and listener prompt the system uses in deployment. No leader: replies are joined and graded as they are.

Questions were written by Claude subagents in three stages. Readers listed 284 facts with verbatim quotes, all confirmed by script. Writers turned them into questions in other words than the source. Checkers that had not seen the writing verified each answer against its quote, searched the eight sections for other passages answering the same question, and fixed 69 items and dropped 9, including one wrong answer and two places where the handbook contradicts itself. The final set is 96 questions and 20 overheard conversations, 15 of them with a relevant fact and 5 with none. GPT-5.4 graded each reply against the accepted answers and source quotes, alongside a text match.

Kind	n	DeepSeek-V4.1-Flash	GLM-5.3-Flash
Answer held by one listener	45	45	45
More than one right answer	8	8	8
Answer changed over time	10	10	10
Answer spread over two or three	19	17	18
No answer in the eight sections	14	12	12
Overheard conversation	20	19	18
Total	116	111	111
Cost of the full run		\$0.59	\$4.16

Table 7: Handbook test, one round, the deployed listener prompt. Judge and text match disagree on one item per listener.

Both listeners answer 111 of 116; DeepSeek does it at a seventh of the cost and is the system’s default listener as a result. The two no-answer items both missed point to the handbook section that holds the answer, which the grader counts as answering. The remaining misses are a needed part left out, because the listener holding it stayed silent or gave only half of it, and, for GLM, two false alarms on conversations where nothing was relevant.

The first grading scored the same replies 98 and 89 of 116. The judge had read the joined replies as one voice, so a listener writing that its part did not mention something, next to another listener’s correct answer, was scored as a contradiction. Once told the reply is several listeners’ notes, judge and text match agreed on all but one item per listener. This is also the clearest weakness the test shows: about a third of contributions came from listeners that did not hold the answer, often to say so. In deployment the leader filters them, but they are paid for.

3.7 Against retrieval

The same 116 items were run through a standard retrieval pipeline: the eight sections cut into 400-token chunks with 50 tokens of overlap, embedded with OpenAI’s text-embedding-3-large, the top k chunks by cosine similarity passed to the same answering model with an instruction mirroring the listeners’. Same judge. A keyword retriever (BM25) is included as the classical baseline.

Kind	n	Herd	RAG $k=30$	RAG $k=10$	BM25 $k=10$
Answer held by one listener	45	45	43	43	21
More than one right answer	8	8	8	8	1
Answer changed over time	10	10	9	8	3
Answer spread over two or three	19	17	11	5	1
No answer in the eight sections	14	12	13	13	13
Overheard conversation	20	19	11	10	7
Total	116	111	95	87	46
Cost of the full run		\$0.59	\$0.18	\$0.09	\$0.08

Table 8: Handbook test, DeepSeek-V4.1-Flash answering in every column. Retrieval cost includes embedding the corpus once (\$0.12).

Retrieval matches the herd on single lookups at a third of the cost. It falls behind in the two cases the system was built for. On answers spread across sections, the search returned every needed document for only 7 of 19 questions at $k=10$, and 11 of the 14 misses trace to a document never retrieved; tripling k recovers about half the gap. On overheard conversations, where there is no question to search with, more context barely helps: 10 of 20 at $k=10$, 11 at $k=30$, against 19 for the herd.

4 Limitations

n is 10 per cell on RULER; single-sample differences between listeners are not meaningful. The chain comparison uses one partition count (7) and one chain shape.

The RULER instruction is written for that benchmark. The handbook test uses the deployed listener prompt, but no leader and none of the Claude Code integration.

The handbook questions were written and checked by one model family (Claude) and graded by another (GPT-5.4); twenty were spot-checked by hand. The handbook is public and likely in the listeners’ training data. We planned to drop any question a listener answers without the handbook and did not run that check, so some answers may come from memory rather than from the held sections.

We do not know why two listeners fail on chains under both instructions. Our count of how many listeners spoke treats an empty reply as speaking, so the speaker counts in Table 2 are upper bounds, and the count cannot be used to explain those failures.

RULER cannot test the system’s central design choice. Its haystacks have no domain structure, so semantic partitioning is not expressible and the results measure *arbitrary* partitioning, a lower bound. Its needles also share vocabulary with the question, which is the easy case; NoLiMa [6] shows accuracy falling sharply once that overlap is removed. Whether domain-coherent partitions do better is still untested; the handbook test, whose partitions are real subjects, makes the comparison possible but it has not been run.

Retrieval was tested in one standard configuration; a reranker, hybrid search or query rewriting could narrow the gap, and were not tried.

The conversational setting that motivates the system is tested only by 20 overheard conversations. Whether a listener recognises relevance when *no question was asked* is the claim the system rests on and still the least tested here.

5 Related work

Closest is ExtAgents [2], which partitions external knowledge across agents and broadcasts a query, but instantiates agents per query rather than persisting them. LLM-based multi-agent blackboard systems [3] have partition-owning agents that volunteer to a central agent, but retrieve on demand rather than holding knowledge resident. Cache-Augmented Generation [4] argues for a resident corpus in a single context, and LOFT [7] benchmarks whether long-context models can subsume retrieval outright. Overhearing agents [5] survey single assistants that monitor ambient activity and intervene when useful, and identify “when to speak” as open.

The contribution here is the persistence-via-cache economics, the measurement that cheap listeners suffice, the finding that the listener’s instruction, not the partitioning, decides whether facts spread across partitions survive, and a public test of realistic questions over a company handbook, with the questions, their sources and the prompts that produced them.

Code

<https://github.com/Florin-Birgu/elephant-herd>

References

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